

Polygenic overlap when one trait is oligogenic

Eleven trait pairs across thirteen traits, one QC arm.
bipred 0.3.6, ldpred3 0.4.5, numpy 2.4.6, Windows-11 x86-64.
Estimates in `results/estimates.csv`.

The question, and the finding

Genome-wide genetic correlation averages over a million variants. For a trait whose heritability sits in one or two loci, that average is dominated by the $\sim 999,900$ variants doing nothing. The question here is whether bipred's polygenic-overlap parameterisation recovers a relationship that `rg` dilutes to noise.

Lipoprotein(a) is the case that makes it matter: essentially monogenic at *LPA*, a validated causal risk factor for coronary artery disease, and the target of phase-3 lowering trials. If `rg` cannot see that relationship, it is the wrong summary for a whole class of clinically important exposures.

It cannot. Fitted against CAD over 958,848 variants, Lp(a) has a genome-wide `rg` of $+0.0481$, and cross-trait LD Score regression puts the same quantity at -0.0945 (se 0.1161) — pointing the wrong way, and within one standard error of zero either way. The same fit estimates 106 causal variants for Lp(a) against 1,459 for CAD, 49 of them shared, with effects inside that shared component correlating $+0.3644$.

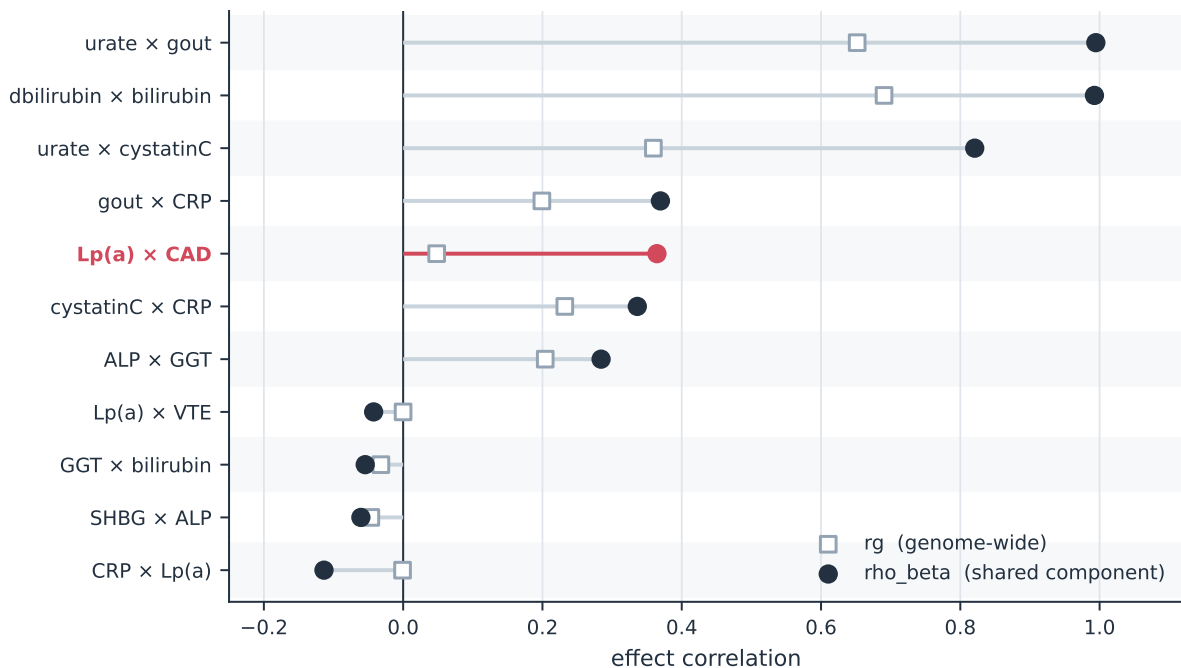


Figure 1: Pairs ordered by `rho_beta`, with genome-wide `rg` on the same row. The order is biological: same-biology and causal pairs at the top, unrelated pairs at the bottom. `rho_beta` exceeds `rg` almost everywhere, because `rg` is attenuated by the variants that carry no signal for either trait. What distinguishes Lp(a) $\times$ CAD is the size of the gap: its `rg` sits with the unrelated pairs while its `rho_beta` sits with the related ones. Compare gout $\times$ CRP directly above it — nearly the same `rho_beta` ($+0.3694$ against $+0.3644$) but four times the `rg` ($+0.1992$ against $+0.0481$).

The two pairs at the top are there because their answers were known in advance: direct against total bilirubin is two studies of the same biology, and urate against gout has crystal deposition as the disease mechanism. Both land near $+0.99$, which is what fixes the top of the scale. Four pairs sit below zero, between -0.11 and -0.04 — three chosen because no shared biology was expected ($\text{GGT} \times \text{bilirubin}$, $\text{SHBG} \times \text{ALP}$, $\text{CRP} \times \text{Lp(a)}$), plus $\text{Lp(a)} \times \text{VTE}$. Without both ends of that scale, $+0.3644$ would be a number with nothing to compare it to.

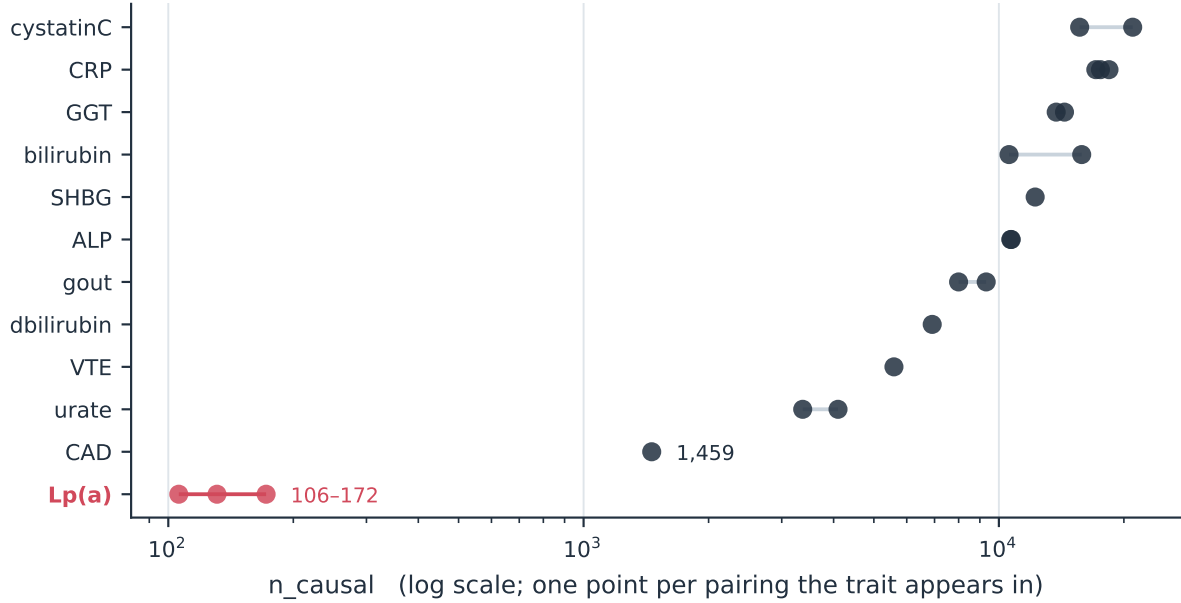


Figure 2: Estimated polygenicity per trait, one point per pairing the trait appears in. Lp(a) is estimated at 106, 131 and 172 causal variants across three independent pairings; every other trait here is in the thousands to tens of thousands. Averaging ~ 100 causal variants over $\sim 10^6$ destroys the signal in **rg**; the architecture summary does not average it away. Note what these counts are measured on: the per-variant QC caps chi-square at 80, which removes 73% of the *LPA* locus before anything is fitted. See *The chi-square cap* below.

frac_shared is not the informative statistic

A second result fell out of the pairs included only to fix the null: sharing causal *variants* is nearly free, and carries little information on its own.

$\text{GGT} \times \text{bilirubin}$ and $\text{SHBG} \times \text{ALP}$ were chosen as unrelated pairs of polygenic traits. They share 63% and 71% of the smaller causal set — and their shared effects correlate at -0.05 and -0.06 . Sitting at nearly the same **frac_shared**, $\text{gout} \times \text{CRP}$ and $\text{cystatinC} \times \text{CRP}$ correlate at $+0.37$ and $+0.34$. For traits that are each causal at tens of thousands of variants, substantial overlap of the causal *sets* is close to the default, and it is the *direction* of the shared effects that discriminates.

The qualification matters: over the whole range **frac_shared** is not uninformative — it runs from 0.39 at the bottom of the scale to 1.00 at the top. What it cannot do is separate pairs in the middle, which is where most real pairs fall.

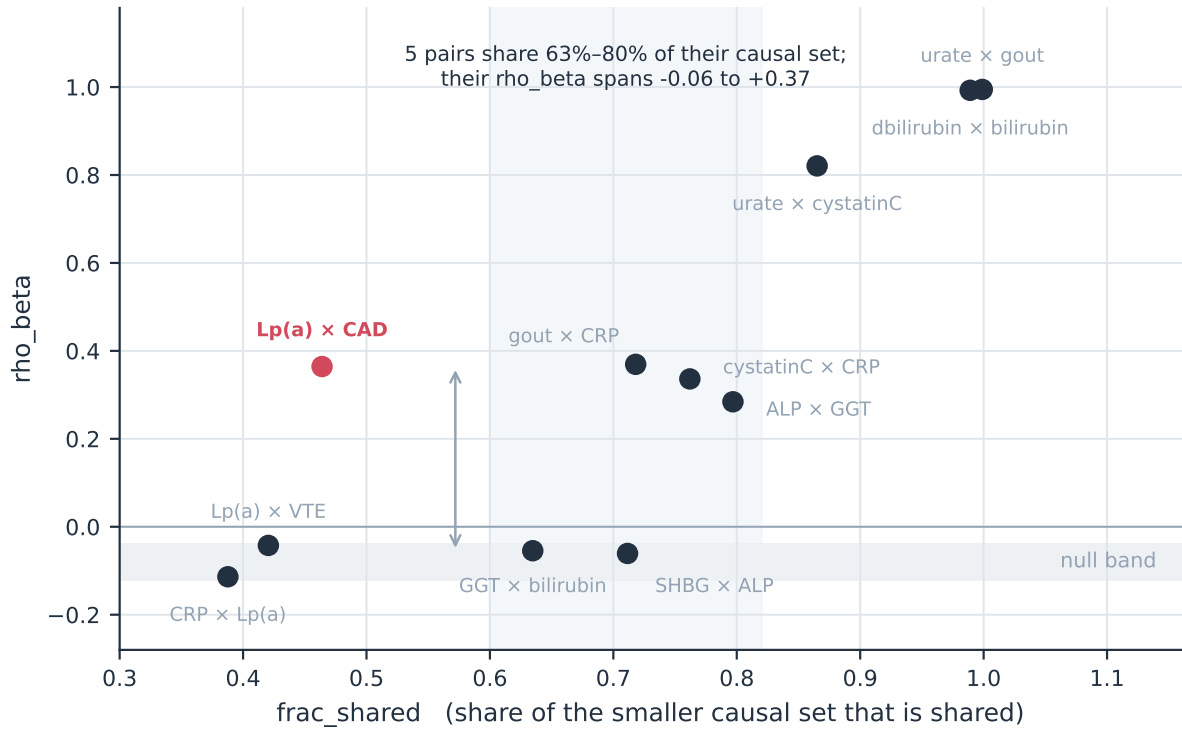


Figure 3: `frac_shared` against `rho_beta`. Across the full range the two are plainly associated — it is the middle that cannot be resolved. Inside the shaded column, five pairs share 63–80% of their causal set while their `rho_beta` spans -0.06 to $+0.37$. `Lp(a) × CAD` sits low on the horizontal axis because `Lp(a)` has few causal variants at all, which is the point: a small shared set with a consistent direction. The horizontal band marks where the four below-zero pairs fall; it is descriptive, not a test.

This matters for how the overlap summary should be read. `n_shared` and `frac_shared` describe how much of the architecture is common; `rho_beta` describes whether the common part agrees. Only the second distinguishes the pairs here.

Agreement with LD Score regression

Both reports lean on cross-trait LDSC as the independent check on the joint fit, so it is worth showing what that check actually looks like across every pair rather than quoting it one pair at a time.

Three further pairs are drawn in from `benchmarks/qc_factorial.csv` at the same arm 001 — `LDL × CAD`, `height × LDL` and `HDL × TG`. They extend the range in both directions: `HDL × TG` is the only strongly negative `rg` available, and `height` is the most heritable trait in either set. They are marked with triangles because they come from a different script and a separately screened variant set; the factorial’s `LDL × CAD` retains 832,668 variants where the `ldl-cad` study retains 845,623, so the two are close but not the same fit.

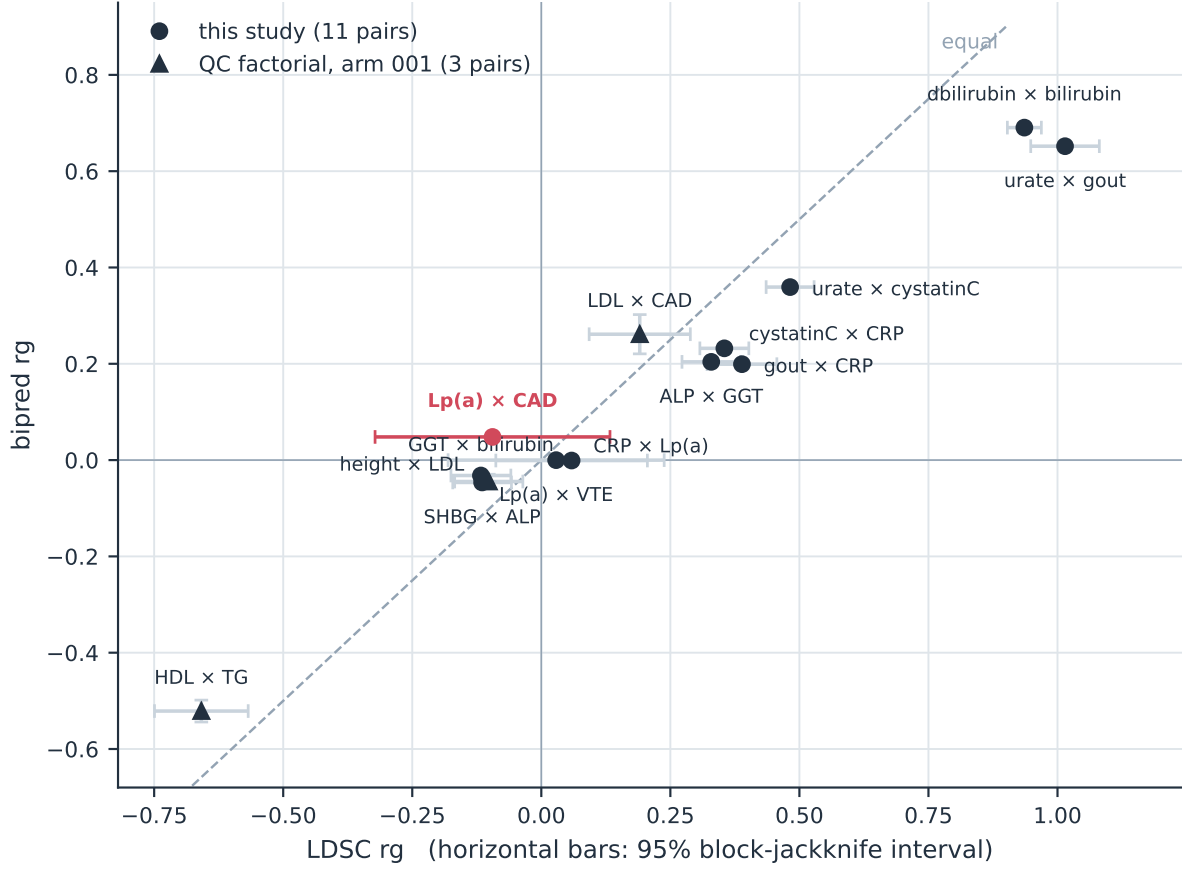


Figure 4: bipred rg against cross-trait LDSC rg . The two agree on sign and ordering across the full range, from -0.66 to $+1.01$. They do not agree on magnitude: bipred is consistently pulled toward zero at both ends — $+0.65$ against LDSC’s $+1.01$ for urate $\times$ gout, -0.52 against -0.66 for HDL $\times$ TG. Lp(a) $\times$ CAD is the pair where they disagree in sign, and it is also the pair with the widest LDSC interval: the jackknife bar runs from -0.32 to $+0.13$, so LDSC is not resolving this pair at all.

That last point is worth separating from the study’s main claim. The reason to prefer `rho_beta` for Lp(a) $\times$ CAD is not that LDSC gives a different answer — LDSC gives an answer with an interval four times wider than the estimate, which is what a genome-wide average does on a trait with ~ 100 causal variants.

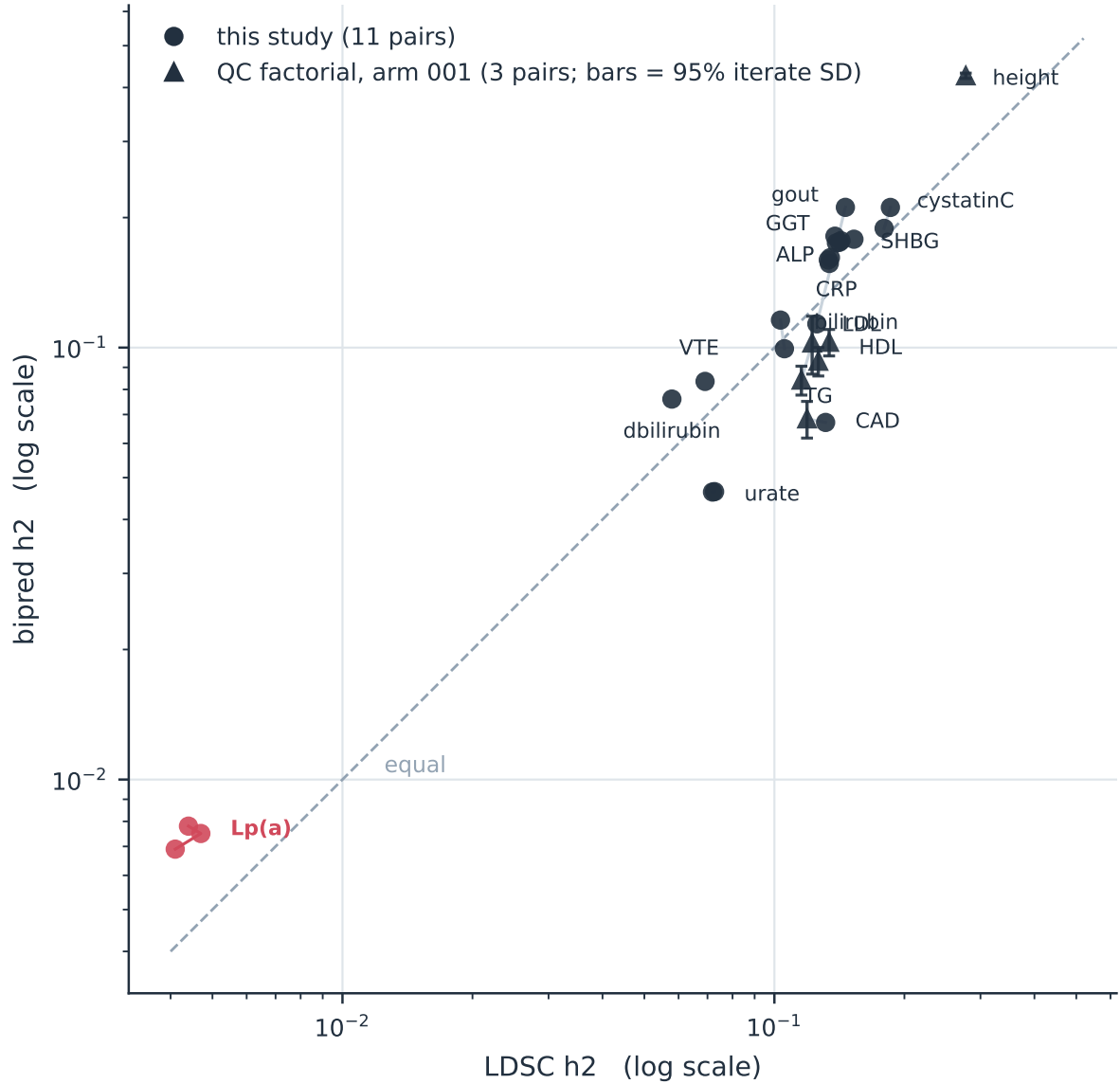


Figure 5: Per-trait h^2 , both estimators, log scale. Points are one trait in one pairing, joined where a trait appears more than once; the joins are therefore an empirical stability check, and most are short. The two estimators track each other over two orders of magnitude, with bipred running slightly high through the middle of the range. Three disagreements stand out: CAD ($0.13 \rightarrow 0.07$) and urate ($0.07 \rightarrow 0.05$), where bipred is lower, and height ($0.28 \rightarrow 0.43$), where it is much higher.

The joins also expose one instability worth recording. Gout moves from h^2 0.11 in urate $\times$ gout to 0.21 in gout $\times$ CRP, an 86% change, while LDSC moves $0.13 \rightarrow 0.15$ over the same two fits. The variant sets differ between pairs, since the screen masks are intersected per pair, but that is a 3% difference in m against a doubling in h^2 . No other trait behaves this way.

What is missing here

Only one of the four quantities plotted above has an interval, and it is LDSC's. That is a recording gap rather than a methodological one:

- **LDSC `rg`** — block-jackknife standard error, recorded everywhere. These are the horizontal bars.
- **bipred `rg` and `h2`** — the factorial records an iterate SD, the vertical bars on the three triangles. This study’s CSV records none. Note what the factorial’s bars measure: spread across retained Gibbs iterates, which is Monte-Carlo variability of the fit and not sampling error of the GWAS. At around 1% of `h2` they are far too narrow to read as confidence intervals.
- **LDSC `h2`** — no standard error at all. `LDSCrgResult` returns `h2` as a bare tuple.

`BivariateResult` retains `genetic_samples`, the raw (`gvar1`, `gcov`, `gvar2`) quadratics at every post-burn-in sweep, so posterior spread for both `h2` and `rg` is recoverable from a fit — the drivers simply do not summarise it into the CSV. Recording those percentiles, and a jackknife standard error for LDSC `h2`, is the change that would let this section carry real intervals. Until then the bars above should be read as what they are.

The chi-square cap

The per-variant QC in arm 001 caps chi-square at 80. That single filter turns out to matter more for this study than any other, and it was found by noticing that `Lp(a)`’s `h2` of 0.007 is too small even for a tagging problem.

Measured on the published `Lp(a)` release (9,183,329 variants with finite β and standard error):

variants above $\chi^2 = 80$	3,819	(0.042%)
share of the trait’s summed χ^2 they carry		42.5%
variants in the <i>LPA</i> window, chr6:160.0–161.1 Mb		3,970
of those, removed by the cap	2,894	(72.9%)
largest, rs10455872		$\chi^2 = 90,550$

Four ten-thousandths of the variants carry 42% of the signal, and the cap deletes them. rs10455872 exceeds the threshold by a factor of 1,132.

Why the cap is there

It is inherited from LD Score regression, and `bipred.ldsc.ldsc_rg` says so directly: it is a one-step unfiltered LDSC whose regression weights are built from the fitted means, so a variant whose observed chi-square far exceeds the LDSC line keeps near-full leverage on the slope. The docstring instructs the caller to cap extreme chi-square before calling it, and the reference implementation excludes $\chi^2 > \max(0.001N, 80)$ for the same reason.

The cause is statistical, not numerical. At $N = 284,044$ a χ^2 of 90,550 is a standardised $\hat{\beta}$ of about 0.56, well inside the $|\hat{\beta}| < 1$ the standardisation requires and nowhere near a precision limit.

Why it should not apply to the joint fit

Both drivers build *one* per-variant mask and pass the same variant set to LD Score regression and to the bivariate fit. So a filter that exists to protect a linear regression also removes those

variants from a Bayesian mixture model whose slab component exists precisely to hold large effects. bipred does not need the cap and pays its cost anyway.

It is not confined to Lp(a)

The obvious guess is that this is an oligogenic-trait problem and that polygenic traits pay almost nothing. That guess is wrong. `../_lib/chi2_cap_audit.py` recomputes the loss for every registered trait, counting only variants present in the LD reference — the only ones a fit could have seen:

trait	over cap	% of $\sum \chi^2$ removed	largest χ^2	at
bilirubin, total	2,262	73.0	107,799	rs887829
bilirubin, direct	788	69.5	76,937	rs887829
lipoprotein(a)	529	50.5	12,694	rs3103347
urate	1,645	46.7	14,649	rs7675964
cystatin C	2,039	27.0	21,274	rs35610040
ALP	2,960	25.5	10,369	rs651007
SHBG	1,957	22.9	8,332	rs727428
CRP	1,268	17.7	5,461	rs4420638
GGT	1,838	16.0	7,772	rs5760492
LDL	420	7.6	2,013	rs445925
gout	395	6.5	1,907	rs2231142
VTE	153	2.2	758	rs651007
gallstones	133	1.9	1,402	rs4953023
CAD	29	0.4	443	rs2891168

Nine of the fourteen lose more than 15% of their signal; four lose more than 45%. Total bilirubin loses 73%, driven by rs887829 at *UGT1A1* at $\chi^2 = 107,799$. Under 3,000 variants out of a million carry that much of the signal in these traits, and the filter is built to delete exactly them.

The bottom of the table is as informative as the top. CAD loses 0.4% and LDL 7.6%, which is why the `ldl-cad` study — the one place this pipeline had been examined closely — could not have surfaced this. Those two traits are nearly the least affected of the fourteen.

This has an awkward consequence for the calibration in the first figure. `dbilirubin` $\times$ `bilirubin` is one of the two pairs used to fix what a large `rho_beta` looks like, at +0.9926 — and both of its traits are the worst affected in the table. The anchor is not damaged in a way that obviously biases it, since both traits lose the same locus and the surviving effects still agree, but it is not the clean reference it was presented as.

What it does and does not change

It does not overturn the ordering in the first figure: `rho_beta` still separates related from unrelated pairs, and it still places `Lp(a)` $\times$ `CAD` with the related ones while `rg` cannot resolve it. Every pair was capped identically, so the comparison between pairs is unaffected.

What it does change is the mechanism. The report cannot claim that averaging one very large locus over a million variants is what defeats `rg` here, because that locus was removed before fitting. The defensible claim is narrower and still worth making: with the dominant locus gone, `Lp(a)`'s residual architecture is around 100 causal variants, `rg` averages that to noise, and `rho_beta` does not. What the uncapped fit would give is unknown, and is the first thing to run.

A prediction the cap makes about polygenicity

There is a second consequence, and it points at numbers already in this report. Truncating the top of the effect-size distribution does not reduce the heritability the sampler has to account for — it removes the large effects that would have accounted for it. The mixture then has to explain the same signal with small effects, and the only way to do that is to use more of them. The cap should therefore *inflate* `n_causal`.

The estimates are high enough to invite that reading: cystatinC at 21,000 causal variants, CRP at 17,117–18,414, GGT at 13,741–14,388, out of roughly 10^6 fitted.

The cross-trait comparison does not support it. Putting the loss column beside `n_causal`, the three traits that lose most — bilirubin at 73%, Lp(a) at 51%, urate at 47% — come in at roughly 13,000, 130 and 3,700 causal variants, which is the full range of the table. The two highest polygenicity estimates, CRP and cystatinC, lose 18% and 27%, less than the median. If the cap were inflating `n_causal`, this is where it would show, and it does not.

That is not a refutation either, because the comparison is confounded: the traits differ in true polygenicity, sample size and how concentrated their architecture is, and those differences are larger than the effect being looked for. The test that would settle it is within-trait — refit the same trait with the cap raised and see whether `n_causal` moves. Until that is run, the mechanism is plausible, the cross-trait evidence is against it, and neither is worth much.

Methods

The model

bipred is bivariate LDpred: one joint fit of two traits that share a single LD reference. Each variant is assigned to one of four latent states — causal for neither trait, for trait 1 only, for trait 2 only, or for *both* — with probabilities $(\pi_{00}, \pi_{10}, \pi_{01}, \pi_{11})$. A trait-1-only effect is drawn from $N(0, s_1)$, a trait-2-only effect from $N(0, s_2)$, and a both-causal pair from $N(0, \Sigma)$ with $\Sigma = \begin{bmatrix} s_1 & s_{12} \\ s_{12} & s_2 \end{bmatrix}$. The off-diagonal s_{12} is the effect covariance inside the shared component, and is the only place the two traits couple.

The indicator is *per trait*, not a single shared one. Whether the two traits’ causal variants co-occur is therefore learned as π_{11} rather than assumed: correlated traits borrow strength through the both-causal component, and disjoint traits can drive π_{11} toward zero so the two fits decouple. That is what makes the polygenic-overlap summary an estimate rather than a restatement of the prior.

Fitting is Gibbs sampling over the LD blocks. Each sweep evaluates the four bivariate-Gaussian likelihoods of the residual marginal estimate at a variant, samples that variant’s state, then draws its effects; the mixture weights π and the variance components (s_1, s_2, s_{12}) are re-estimated every sweep under a damping factor of 0.2. Reported quantities are posterior means over the retained iterations: per-trait SNP heritability `h2`, the genetic correlation `rg`, posterior-mean effects, and a MiXeR-style overlap summary — `n_causal` per trait, `n_shared`, `frac_shared`, and the correlation `rho_beta` of effects within the shared component.

What rho_beta is, and is not

`rg` is a genome-wide average: genetic covariance divided by the geometric mean of the two heritabilities, over every variant in the fit. `rho_beta` is the effect correlation *conditional on a variant being causal for both traits*. The two answer different questions, and they come apart exactly when one trait’s heritability sits in very few loci. Neither is a causal estimate.

`frac_shared` is π_{11} divided by the smaller of the two marginal causal fractions, so it asks what proportion of the less polygenic trait’s causal set is also causal for the other.

Sample overlap

Overlapping samples correlate the two traits’ sampling errors. The fit accepts that correlation as `cross_corr`, set per pair from the cross-trait LD Score regression intercept, used as-is and never clipped into range. An intercept can also reflect correlated confounding rather than shared samples, so this absorbs overlap without identifying it.

The LD-consistency screen

Per-variant filters — minor allele frequency, imputation quality, a chi-square cap, sample size — judge each variant in isolation, so none of them can see a variant disagreeing with the variants it is correlated with. The screen tests exactly that. Within a window the variants are split at random in half and each z -score in one half is predicted from the other through the LD,

$$\hat{z}_a = R_{aB} R_{BB}^+ z_B, \quad T_a = \frac{(z_a - \hat{z}_a)^2}{1 - R_{aB} R_{BB}^+ R_{Ba}} \sim \chi_1^2$$

under consistency. Variants far from what their neighbours predict are dropped, and the split is repeated so each variant is tested from several directions.

It runs against the same numeric LD representation the fit will use, so the screen and the sampler see the same matrix. It is DENTIST-inspired, not DENTIST: window construction, partition schedule, eigenvalue regularisation and removal policy all differ from the published pipeline (Chen et al. 2021, *Nature Communications* 12:7117), so that pipeline’s calibration does not transfer.

Divergence diagnostics

A bivariate fit can diverge silently. Given summary statistics that disagree with the LD reference, the sampler places large opposing effects on variants in near-perfect LD, and these cancel to a plausible-looking `h2`. Three diagnostics are recorded, and any one of them raises the warning: $\sum \beta^2/h^2$ above 10 (the cancellation ratio, reported in both studies), $\max |\beta|$ above 25 slab standard deviations, and retained-trace drift above 1.25. All three were calibrated on one real LDL $\times$ CAD fit that diverged and the same fit after screening repaired it, so each separates the two regimes by more than an order of magnitude rather than by a hair.

An estimate carrying the divergence warning is not an estimate of anything, and is reported only to show the contrast.

LD reference and harmonisation

All fits use the European UK Biobank HapMap3 LD reference, built with ldpred3's bigsnpr converter at the pinned revision 5d86ac9d. Summary statistics are harmonised to it by rsID with allele matching, flipping the effect sign where the effect and reference alleles are swapped; strand-ambiguous and unmatched variants are dropped. Odds-ratio releases are taken to logs before harmonisation, which preserves the sign across a flip. Case/control traits are fitted at an effective N of $4/(1/n_{\text{case}} + 1/n_{\text{ctrl}})$. BLAS is pinned to one thread throughout.

This analysis

Every pair was fitted under a single QC arm, **001**. Its per-variant filters, in the order `benchmarks/qc_factorial.py` applies them, are reference $\text{MAF} \geq 0.01$; allele-frequency discordance against the reference ≤ 0.20 ; imputation $\text{INFO} \geq 0.9$; $\chi^2 \leq 80$; per-variant $N \geq 0.67 \times \text{median}(N)$; and an implied-standard-deviation consistency check at the lenient bounds (0.5, 0.1). Long-range LD regions and the MHC are *retained*, and the LD-consistency screen is *on* at four rounds. That arm was chosen because the companion `ld1-cad` study found the screen to be the load-bearing switch of the three; fixing the other two isolates it.

The $\chi^2 \leq 80$ term in that list is not the innocuous filter it looks like, and it has its own section below.

The screen is run once per trait on the lenient variant set, and the resulting masks are intersected across the pair — not re-run per pair. Re-screening per pair would let the screen see a different variant set each time, which entangles what the per-variant filters did with what the screen was given.

Traits, source files, sample sizes, ancestries and known dataset defects are registered in `../_lib/traits.toml`. Sources that were tried and rejected are recorded with their symptoms in `../_lib/datasets.md`; read that before adding a dataset, because two of the three defects recorded there are invisible until a fit has already started.

This is one QC arm on one LD reference. It measures how the two summaries behave across these pairs; it is not a comparison of QC strategies, and the factorial that would support one is in `benchmarks/qc_factorial.py`.

Full results

The figures above are a reading of one table, not a substitute for it.

<code>results/estimates.csv</code>	one row per pair: <code>m</code> , LDSC and bipred <code>rg</code> , <code>h2</code> , the overlap parameters, cancellation ratios, divergence flag
<code>results/provenance.json</code>	package versions, platform, clean source revisions
<code>../_lib/traits.toml</code>	every trait: file, column layout, N , ancestry, known quirks
<code>../_lib/datasets.md</code>	sources tried and rejected, with symptoms

Thirteen rows are recorded: the eleven fitted pairs above, and two fits of bilirubin $\times$ gallstones that were rejected on data grounds rather than on their results. The gallstones release carries minor allele frequency under an effect-allele-frequency header, and the refit diverged (cancellation

14.29). Both rows are kept in the CSV so the rejection is auditable; `../_lib/datasets.md` records how it was diagnosed.

Summary statistics are not committed — roughly 9 GB across the thirteen traits. Nothing in this report requires them: `figures.py` reads only `results/estimates.csv`.

Caveats

- **The overlap statistics carry no interval.** The null band is descriptive, not a test, so “+0.3644 is outside it” is an observation about eleven fits and not an inference. This is the largest gap in the analysis, and it is a recording gap: `BivariateResult.genetic_samples` retains the per-sweep quadratics the intervals would come from, but the drivers summarise only their means into `results/estimates.csv`.
- **$rg = +0.0481$ for $Lp(a) \times CAD$ is not a null result**, and will be read as one if it is quoted without the architecture that explains it.
- **Every oligogenic reading here is $Lp(a)$.** Its `n_causal` reproduces across three pairings (106, 131, 172), which is reassuring about the estimate but says nothing about whether the effect generalises to other oligogenic traits.
- **$Lp(a)$ ’s h^2 of 0.007 is mostly a QC artifact.** The chi-square cap of 80 removes 73% of the *LPA* locus, including rs10455872 at $\chi^2 = 90,550$, before either estimator sees the data. Poor HapMap3 tagging of the KIV-2 repeat is a second, smaller cause. It is not an estimate of $Lp(a)$ heritability. See *The chi-square cap*.
- **$Lp(a)$ is a four-cohort meta-analysis at 95.9% European** fitted against a European LD reference. The European-only alternative covered 16% of the reference and was unusable.
- **Sample overlap is substantial** — most exposures are UK Biobank. The cross-trait LDSC intercept absorbs it and is recorded per pair, but an intercept cannot identify overlap on its own.
- **The Sinnott-Armstrong panel publishes no allele frequency**, so the AF-concordance check is vacuous for those traits and the +1.0000 it reports is an artifact of comparing a column with itself.